Tesseract: Interactive Conversion of Building Floor Plans into Multi-Floor Navigable Graphs [Demo]

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Abstract

Converting architectural floor plans into machine readable navigable graphs is a prerequisite for indoor routing, evacuation analysis, and spatial reasoning, yet it usually demands manual modeling or specialized capture hardware. We demonstrate Tesseract, an interactive web system that turns an annotated building floor plan image into a typed navigable graph in real time. A visitor uploads a floor plan or selects a curated example and watches the pipeline run live through text detection, label interpretation, flood fill segmentation, door detection and classification, connectivity construction, and pruning. The resulting graph is explored on an interactive canvas with node type filters, an adjustable floor plan overlay, and a pre versus post pruning view, and a live connectivity check flags any room or exit that routing cannot reach. The visitor can repair extraction defects by hand, with every edit undoable back to the originally extracted graph, and can export the result as an image or as JSON. The centerpiece is live indoor navigation, where the visitor picks a start node and a destination node and the system animates the shortest route found by in browser A* search, climbing through stair and elevator transitions when the route crosses floors. Tesseract runs in the browser against a deployed backend and is publicly available, so attendees can test it on their own floor plans.

CCS Concepts

• **Information systems** → **Geographic information systems**; • **Computing methodologies** → *Computer vision*; • **Theory of computation** → *Graph algorithms analysis.*

Keywords

floor plan parsing, indoor navigation, navigable graphs, interactive visualization, graph editing, shortest path, demonstration

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1 Introduction

Indoor navigation, evacuation planning, facility management, and embodied artificial intelligence all depend on structured representations of building interiors that encode not only geometry but also the connectivity between rooms, corridors, and floors. Unlike outdoor environments, where map data is abundant, most buildings have no machine readable model of their navigable space. The information needed to build one is usually locked inside a static floor plan image.

Floor plan images are widely available, yet turning them into navigable graphs remains hard. Floor plan parsers such as FloorNet [16] and CubiCasa [11] reconstruct walls and room polygons, but they target geometry rather than traversability, and they rarely address vertical connectivity across floors. Standards such as IndoorGML [15] define rich navigable models, yet populating them still requires manual annotation. As a result, navigable indoor graphs stay expensive to produce at scale.

We demonstrate Tesseract, an interactive system that closes this gap. The Tesseract pipeline [1] contributed the underlying floor plan to graph construction, and Tesseract turns it into a deployed web application that any conference attendee can drive, inspect, repair, and route over. From a single annotated floor plan image, the system produces a typed graph whose nodes are rooms, corridors, outdoor regions, doors, and stair or elevator transitions, and whose edges encode walkable connections weighted by distance. The interface lets a visitor run the pipeline live, watch every stage of the construction, diagnose and fix defects in the result, and route between any two nodes over the graph entirely in the browser.

This demonstration makes three contributions.

- An interactive, publicly deployed system that converts a floor plan image into a typed navigable graph and streams every pipeline stage to the user in real time.
- A graph exploration and repair interface that exposes the internal structure of the construction through node type filtering, an adjustable floor plan overlay, a pre versus post pruning comparison, a live connectivity diagnosis, and hand editing that is undoable back to the originally extracted graph.
- Live in browser navigation, where the user selects a start and a destination and the system runs A* over the constructed graph and animates the route, climbing through stair and elevator transitions when the path crosses floors.

The running system is available at https://huggingface.co/spaces/yansari/Tesseract.

2 Related Work

Most prior work on floor plans reconstructs geometry rather than the connectivity an agent needs to move through a building. Early systems vectorize walls and rooms from raster drawings [17, 18, 23], and learning based parsers such as FloorNet [16], CubiCasa [11], and boundary aware segmentation [27] recover room polygons and wall geometry, frequently from three dimensional scans or richly annotated sources. Symbol and door recognition [4, 6] and surveys of floor plan analysis [10] share the same goal. These methods recover what a building looks like rather than how an agent moves through it, and they seldom model connectivity across floors.

A second line of work models traversability directly. Standards such as IndoorGML [15] and its comparison with CityGML [20], together with building information models [9, 24], define rich navigable representations, yet populating them still demands manual modeling or instrumented capture. In robotics, metric topological and spatial semantic maps [12, 13, 22] and recent scene graph perception [8] produce navigable graphs by physically traversing a space with sensors [7, 25]. This is accurate but does not scale to the many buildings whose only record is a drawing.

Vision language models can now read floor plan maps [5] and reason over top view layouts [14, 26], yet they target question answering rather than a routable graph and are not offered as interactive tools. Crowdsourced and tagging systems add semantics to plans through human effort [21, 28]. To our knowledge no public system takes a single annotated floor plan image to a typed, multi floor, routable graph that a user can inspect, repair, and navigate in the browser. Tesseract closes this gap, and the demonstration makes every step of the construction visible.

3 System Overview

Tesseract has two parts, an image processing pipeline that converts a floor plan into a typed navigable graph, and a web application that serves that graph and makes it explorable and routable in the browser. We describe the pipeline first, then the web architecture that delivers it.

3.1 From Image to Navigable Graph

Tesseract processes a floor plan in a sequence of loosely coupled stages, shown in Figure 1. A CRAFT detector [3] localizes text, and a VGG-BiLSTM-CTC recognizer [2] reads it. A semantic interpreter sorts each label into a room, a corridor, an outdoor region, or a vertical transition such as a stairwell or an elevator. Flood fill segmentation grows each labeled region from its text seed, and a distance transform places interior subnodes that capture room extent for fine grained routing. A Faster R-CNN detector [19] finds doors and classifies them by the regions they connect, such as room to corridor, room to room, corridor to corridor, and exit. The graph constructor links these elements into a typed graph and funnels each room family through a main node so that every interior point reaches every door, then a pruning step removes edges that no shortest path needs while preserving connectivity. For a multi floor building, the per floor graphs are merged and stitched together at validated stair and elevator transitions. On a three floor university building of 344 rooms, the underlying pipeline reaches 94% intra floor navigational completeness and builds the whole building in about three and a half minutes, and pruning removes the bulk of the redundant routing nodes and edges while preserving connectivity.

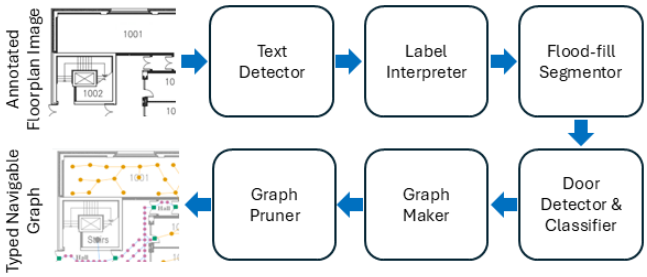


Figure 1: The Tesseract pipeline. An annotated floor plan image is taken through text detection and interpretation, flood fill segmentation, door detection and classification, graph construction with room family funneling, and pruning, yielding a typed navigable graph that the web frontend renders and routes over.

3.2 Interactive Web Architecture

The system is delivered as a single deployable application. A FastAPI backend wraps the pipeline and serves both freshly processed uploads and a set of precomputed example graphs for instant response. A React and TypeScript frontend renders the graph with Cytoscape.js and drives all interaction in the browser. The application is packaged as a Docker image and deployed as a Hugging Face Space that runs on CPU, which keeps the demonstration self contained and reproducible. Because the graph and its edge weights live in the browser, routing queries run locally with Cytoscape A* and return with no server round trip.

4 Demonstration

The demonstration is organized as three short interactions that a visitor performs at the booth. Each one builds on the previous, and together they take a floor plan from a flat image to a building a user can navigate.

4.1 Building a Graph Live

A visitor begins by dropping a floor plan image onto the upload panel or by picking one of four curated examples. An upload runs the full pipeline on the backend, and a live indicator reports the current stage as it advances from detecting text regions, through interpreting labels, flood filling rooms, detecting and classifying doors, building room and corridor connectivity, and funneling room paths to doors, to pruning the graph. Within a few moments the typed graph appears, drawn over a faint copy of the original floor plan so the visitor sees how each node maps to a real room or corridor (Figure 2). The four curated examples return instantly from a result cache, which keeps the booth responsive even while a fresh upload is still processing.

4.2 Exploring and Repairing the Construction

Once a graph is built, the visitor inspects how it is put together and repairs any defects it finds. Node type filters toggle rooms, doors,

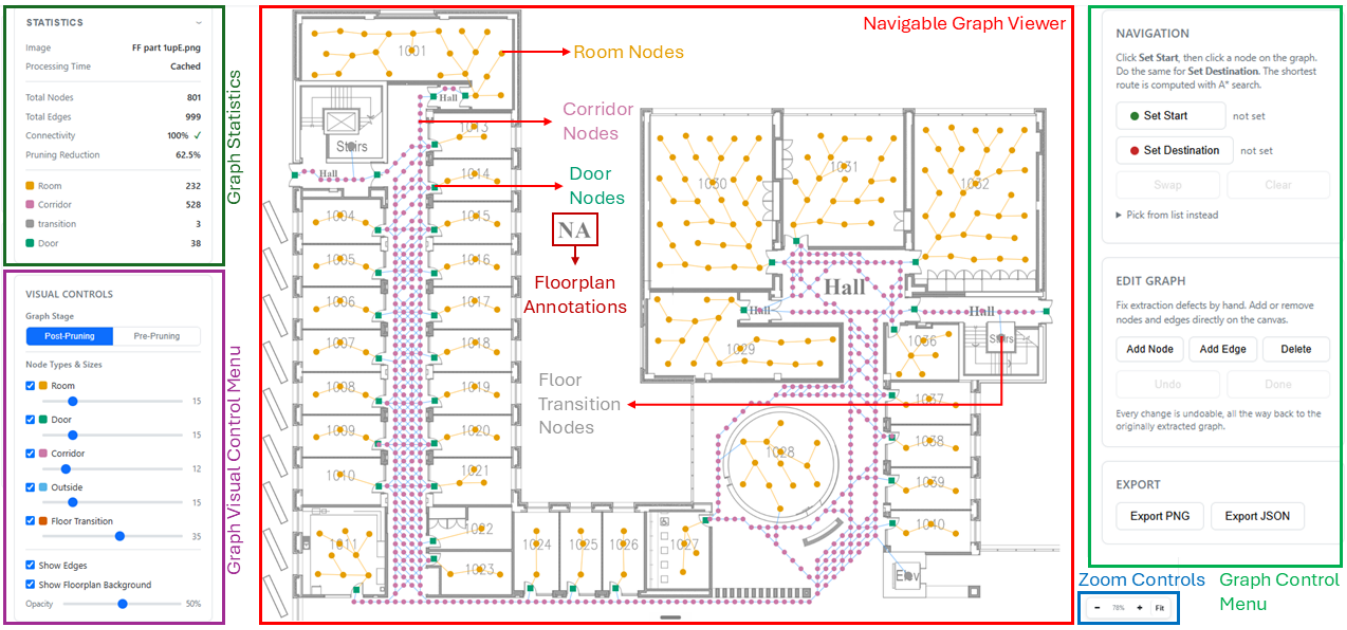


Figure 2: An annotated overview of the Tesseract interface. The constructed graph is drawn over a faint copy of the floor plan in the navigable graph viewer, with room, corridor, door, and floor transition nodes. The visual control menu on the left toggles node types, sizes, and the overlay, the graph controls menu on the right drives navigation, editing, and export, and the zoom controls sit at the lower right.

Table 1: Sizes of the navigable graphs Tesseract builds for the four curated examples in the running demo, where FF and SF denote first and second floor plans. The node total counts room subdivisions and corridor routing points in addition to the labeled rooms, corridors, doors, and transitions.

Example	Rooms	Doors	Nodes	Edges
FF-1	34	38	801	999
FF-2	35	43	682	865
FF-3	25	31	458	501
SF-1	34	38	875	1108

corridors, outdoor regions, and floor transitions independently, each drawn in a colorblind safe color, and a per type size control adjusts how prominently each class is shown. An overlay switch fades the floor plan in and out behind the graph, and an opacity slider tunes how strongly it shows through. The most instructive control flips between the pre pruning and post pruning graphs, so the visitor watches hundreds of redundant edges fall away while the building stays fully connected (Figure 3). Table 1 lists the four curated examples, whose pruned graphs span several hundred to over a thousand nodes while each plan stays fully connected. A statistics panel reports node and edge counts, the pruning reduction, and a per type breakdown, and hovering any node reveals its type, floor, area, and door class. A live connectivity check scores how much of the graph is reachable and flags any room or exit that routing cannot reach, ringing the offending nodes on the canvas so the visitor can jump straight to them. The visitor can then repair these defects by hand, adding or removing nodes and edges directly on the canvas, with every change undoable all the way back to the originally extracted graph. Either graph can be exported as a high resolution image or as JSON for downstream use.

4.3 Navigating the Building

The final interaction turns the graph into a tool. The visitor sets a start node and a destination node, either by clicking the Set Start and Set Destination buttons and then clicking nodes on the canvas, or by choosing them from two menus. As soon as both ends are set, the system runs A* search over the weighted graph, using the straight line distance to the destination as an admissible heuristic, and animates the view to frame the result. The route is highlighted while everything off the path is dimmed (Figure 4), and a short readout reports the path length and the number of segments. The

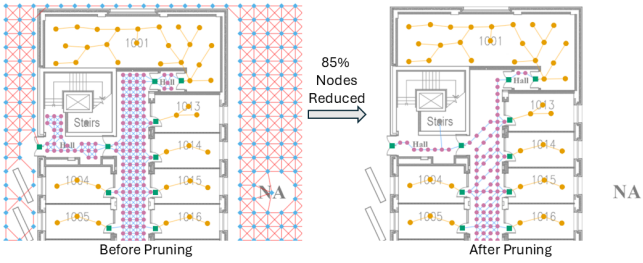


Figure 3: Before and after pruning for one floor. Tesseract removes about 85% of the redundant routing nodes while every room stays reachable.

same query routes from any room to the nearest exit door, which makes the graph immediately useful for evacuation planning. When the start and destination lie on different floors of a stitched multi floor graph, the route climbs through a stair or elevator transition and a Crosses floors badge marks the change of level, which makes the vertical structure of the graph tangible. Because the search runs in the browser, each query resolves in well under a second, so a visitor can try many origin and destination pairs in quick succession.

5 Conclusion

Tesseract makes the construction of navigable indoor graphs visible and interactive. A visitor watches a floor plan become a typed graph, inspects how pruning simplifies that graph without breaking it, repairs any defect by hand, and then routes through the building across floors. The system is deployed and public, and we invite attendees to bring their own floor plans. The running demonstration, a short video, and the source code are linked from the project page.

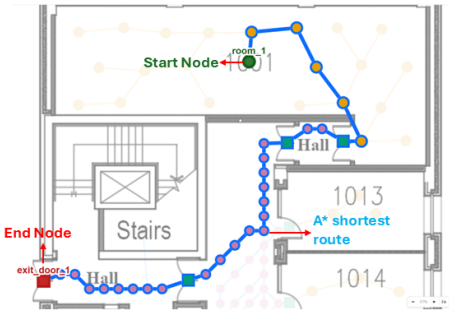


Figure 4: A shortest route computed with A*, here from a room to an exit door. The path is highlighted and the off route structure is dimmed, so the route through the corridor network and past the stairwell stays legible.

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